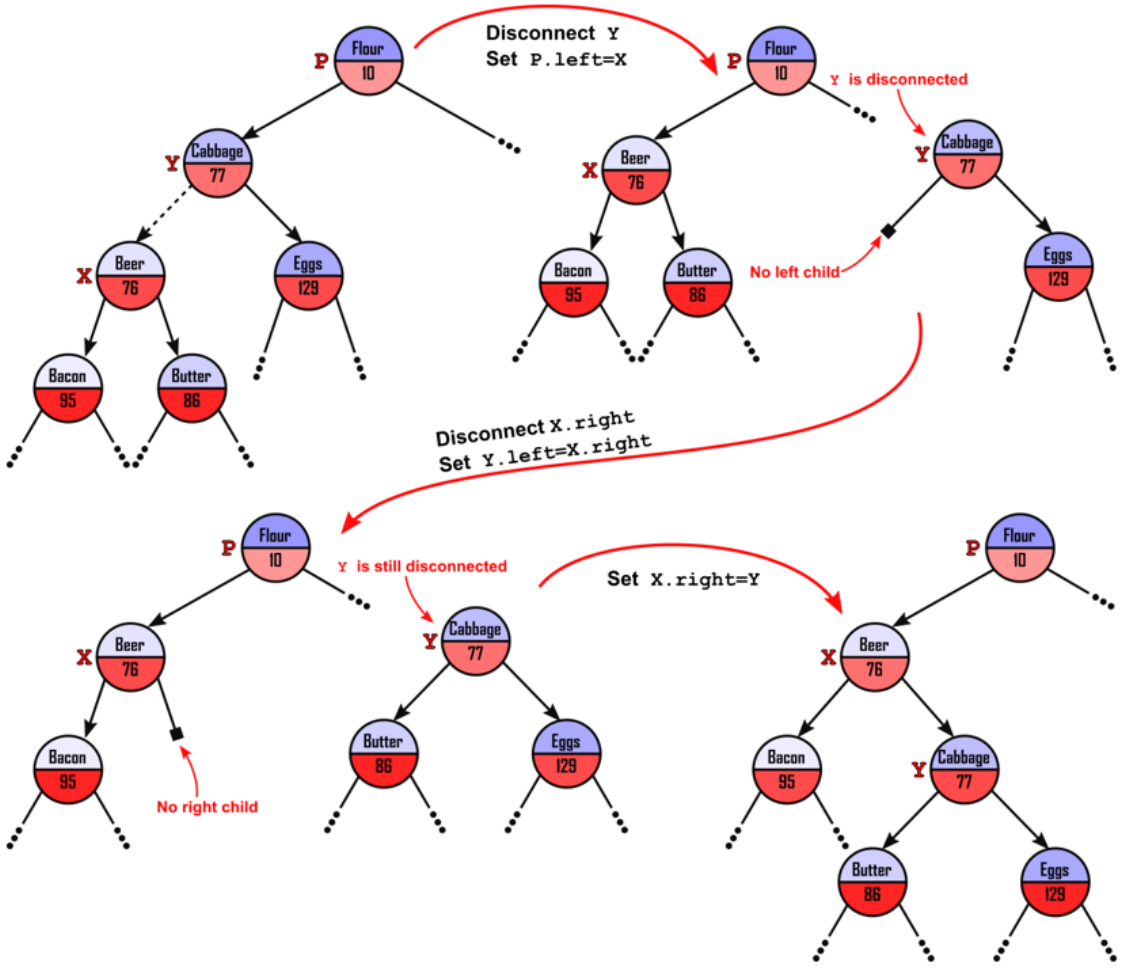


Figure 3.5 The algorithm for performing a right rotation on a BST illustrated.

All that's left to do now is reconnecting Y to the main tree: we can assign it to X's right child (line #11), and we won't have any violation; we already know, in fact, that Y (and its right sub-tree) have larger keys than X's, and for what concern Y's left subtree, it was constructed using the former right subtree of X, and hence by definition all those keys too are larger than X's.

We saw how to perform a rotation: nothing fancy, it's just about updating a few links in the tree… the only mystery at this point might be, why is it called a rotation?